

The Simulation Segment

Plain-words version · v2 of this document, with your answers from 14 Sep built in · SIH PS 26025

In one line

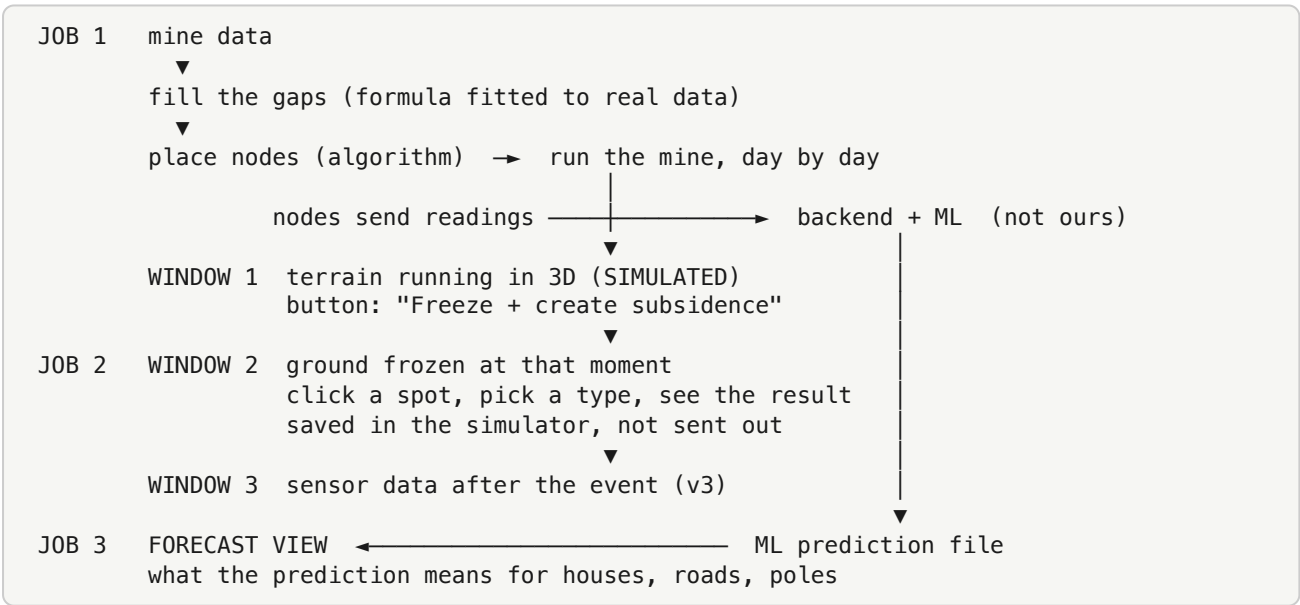
The simulator is a **virtual mine**. You give it a mine's data. It fills the gaps, places the sensor nodes and runs the mine. Then it lets you **freeze the ground and create subsidence** to see what happens, and it shows **what an ML prediction means on the ground**.

It **generates, simulates and displays**. It does not predict (ML does that), and it does not alarm, send SMS, sync offline or push app alerts (backend does those).

1. Who does what

Part	Owner	Job
Simulation	Adarsh + Antigravity	Jobs 1, 2, 3 below
ML	ML teammate	Predicts future sinking from the node data
Backend	Backend pair	Stores data · alarm rule · SMS / app / email alerts · offline mode
Hardware	Hardware pair	The real sensor node
Dashboard	Frontend (owner TBD)	Screens for users

2. The whole thing in one picture



3. Job 1 · Virtual mine (v1, Tue 15 Sep)

1. **Mine data in:** Adriyala mine. Panel 250 m × 2500 m, 375 m deep, 283 real measured points.
2. **Fill the gaps** with a formula fitted to that real data (tilt, stretch, values between survey dates, sensor noise, radio loss). Everything filled in is tagged **synthetic**.
3. **Place nodes** with the algorithm, based on the mine's shape and where the ground moves.

4. **Run** 690 days. The coal face moves and the ground sinks.
5. **Nodes send data forward**: a data file plus a live stream, some packets lost.
6. **Window 1**: a 3D view of the ground sinking, with nodes, a day slider and a **SIMULATED** banner.

4. Job 2 · Freeze and create subsidence (v2, Wed 16 Sep)

How it works: in Window 1 you press **Freeze** at any day. Window 2 opens with the ground exactly as it is at that moment, and Window 1 keeps playing back the finished run, untouched. You click a spot and pick a type. The simulator reads the **data near that spot at that moment** and works out what happens: how much the ground there has already sunk, how much it can still sink, the depth, and the seam height.

#	Scenario type	What decides the result (the "nearby data")	When
1	Sudden local sinking	The sinking that hasn't happened yet at that spot happens all at once. Over old settled ground there's little left, so the result is small. Just behind the face there's a lot, so the result is big. Over solid coal there's no empty space, so it's "not possible here". Depth sets how wide it spreads.	Wed · first batch
2	Ground cracking	How fast the ground near the spot is moving now (last 24 h). Carry that on for N days, then show where the stretch is big enough to crack, and how wide.	Wed · first batch
3	Tilt (edge failure)	The coal pillar at the nearest panel edge gives way, so the sunken area gets wider on that side. The ground next to the edge tilts sharply.	Wed · first batch
4	Blast vibration	Distance from the blast and charge weight give shaking speed (mm/s) at every point, checked against Indian DGMS limits. It needs two site constants from a published Indian study; until we find them, it says "not set" instead of guessing.	Wed · second batch
5	Sinkhole (old shallow workings)	Void height, how much broken rock swells, and the cover depth. If the swollen rock fills the void first, the answer is "no sinkhole reaches the surface" and says why. (Sinkholes caused by water washing soil into old workings aren't modelled; the screen says so.)	Wed · second batch

Every scenario shows the same results: extra sinking · tilt · stretch/squeeze (strain) · cracks with width · effect on each object on the ground · shaking, for blasts only. A plain-words summary sits on top.

Scenarios are labelled **SCENARIO**. They are **stored in the simulation's own storage** (a list of saved scenarios) and **never sent to the backend or ML**.

5. Job 3 · What a prediction means (v2, Wed 16 Sep)

We don't build the ML. We build the **pipe**: prediction in, meaning out.

```

ML file (per node: sinking at +24 h and +72 h, low / likely / high)
  | check the file is valid
  ▼
spread the predicted change from the nodes over the ground → add it to today's ground
  ▼
tilt · strain · cracks · objects → FORECAST VIEW, with a low / likely / high switch

```

Until the ML teammate's real file arrives, a **test file** built from the simulator is used. The screen says **TEST INPUT, NOT A REAL FORECAST**.

6. Objects on the ground (from research on Indian coalfields)

Coal areas in India (Jharia, Raniganj, Godavari valley) have villages, roads, railway lines, power lines and water pipes directly above the workings. The Dhanbad–Chandrapura railway line was shut in June 2017 because of the underground fire and subsidence risk in Jharia; it reopened in 2019 after local protests.

Object	What we show	Limit used	When
House (brick / mud)	Damage grade I–IV from tilt, strain, curvature · shaking check	Published building damage grades for mining areas · DGMS vibration limits	Wed · first batch
Village road	Crack width across the road · tilt along it	Numbers only (no sourced limit yet)	Wed · first batch
Power pole / tower	How far the top leans (cm)	Numbers only	Wed · first batch
Railway track	Uneven sinking under the rails	Numbers only	Wed · second batch
Buried water pipe	Stretch along the pipe	Numbers only	Wed · second batch

We only give a grade where a published limit exists. Everywhere else we show the physical number in plain words, e.g. "the top of the 10 m pole moves 12 cm". The objects are placed as an **example village**, not the real surface above Adriyala.

7. Rules that never change

- **Fake is never shown as real.** Banners: SIMULATED SCENARIO FORECAST.
- A scenario **never changes the main run**. Window 1 and the data file sent to the team stay exactly the same.
- The simulator **never raises an alarm**. Damage grades describe the ground; they don't send alerts.
- Same input gives the same result, every time.
- No numbers typed into code. Every value lives in a settings file with its source.

8. Honest limits

- One real mine dataset. The formula is about 27% low at the deepest point.
- Scenario settings like pillar width, blast charge and old-working depth are your inputs, not measured data from this mine.
- Some limits (roads, poles, rails, pipes) aren't sourced yet, so we show numbers only. Crack **width** is an estimate; the point where cracks start (about 3 mm per metre of stretch) is from published studies.
- Houses near the panel edges already show damage grade II–III before any scenario. That's what longwall mining at this depth does, not a bug.

9. Timeline

When	Done means
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Tue 15 Sep, 20:00 · v1	Job 1 complete · Window 1 working · team demo
Wed 16 Sep, 18:00 · v2 first batch	Window 2 with scenarios 1–3 · Job 3 pipe with the test file · house, road, pole
Wed 16 Sep, 23:00 · v2 complete	+ blast and sinkhole · railway and pipe · real ML file if it has arrived
Later · v3	Window 3 (sensor data after the event) · more scenario types · second mine · detailed radio

The detailed build plan with every prompt is in `work-with-tools/BUILD-PLAN.md`. Your own steps are in `hands-on/MY-STEPS.md`.